

Homework

Preliminary Mathematics Standard

Formulas and Equations

1 Questions

1.1 Formulas

1. Given the formula for the area of a trapezium:

$$A = \frac{h(a+b)}{2}$$

where h is the height, a and b are the lengths of the parallel sides. If $h = 8$, $a = 5$, $b = 12$, find A .

2. The kinetic energy E of an object is given by:

$$E = \frac{1}{2}mv^2$$

Find E when $m = 4.5$ kg and $v = 10$ m/s.

3. In the formula $v = u + at$, find u when $v = 30$, $a = 4$, $t = 5$.
4. The volume of a cone is $V = \frac{1}{3}\pi r^2 h$. If $V = 150$, $r = 5$, find h (take $\pi \approx 3.14$).
5. A car travels 240 km in 3 hours. What is its average speed in km/h?
6. If a car is travelling at 90 km/h, how far will it travel in 20 minutes?
7. Use the formula for stopping distance:

Stopping distance = reaction distance + braking distance

where reaction distance = speed $\times$ 0.75 (in metres, if speed in km/h), and braking distance = $0.008 \times (\text{speed})^2$. Find the total stopping distance for a speed of 60 km/h.

8. The Blood Alcohol Content formula for males is:

$$BAC_{\text{male}} = \frac{10N - 7.5H}{6.8M}$$

where N = number of standard drinks, H = hours drinking, M = mass in kg.

- (a) Calculate the BAC for a male of mass 85 kg who has consumed 6 standard drinks over 3 hours.
- (b) How many hours H are needed for the BAC to reach zero if $N = 8$, $M = 78$ kg?
- (c) Discuss **two** limitations of using this formula to estimate BAC.

9. Clark's rule for child dosage is:

$$\text{Child dose} = \frac{\text{Adult dose} \times \text{Child's weight (kg)}}{70}$$

If the adult dose is 500 mg and the child weighs 25 kg, find the child's dose.

10. Fried's rule for infants under 2 years:

$$\text{Infant dose} = \frac{\text{Adult dose} \times \text{Age (months)}}{150}$$

If the adult dose is 300 mg and the infant is 18 months old, find the infant's dose.

- 11. Make r the subject of $A = \pi r^2$.
- 12. Rearrange $v^2 = u^2 + 2as$ to make u the subject.
- 13. The formula $y = 3x^2 - 12$ can be written as $x^2 = \frac{y+12}{3}$. Solve for x if $y = 33$.
- 14. When parachuting from an airplane, parachutists wait 6 seconds before they pull the ripcord to release the parachute.
 - (a) Use the formula $d = 4.9t^2$, where d is the distance in metres and t is the time in seconds, to calculate the distance that a parachutist falls during the first 6 seconds of a jump.
 - (b) The approximate speed (s) in kilometres per hour that the parachutist travels during free fall (when the parachute is closed) can be found using the formula $s \approx 36t$, where t is the number of seconds after jumping from the plane. Find the approximate speed at which the parachutist is traveling when she pulls the ripcord.
 - (c) The terminal velocity (maximum speed) of a parachutist is about 200 kilometres per hour. Has the parachutist reached terminal velocity at 6 seconds?
- 15. A student was discussing the formula for a hire car company, $C = 0.95D + 30$, where C is the total cost of the car hire and D is the distance travelled in kilometres. The student says that to hire a car from this rental company costs 95 cents a kilometre. Is the student correct? Explain.

16. Stopping distance is the distance required to bring a moving vehicle to a stop from the moment the brakes are applied. The formula to calculate the stopping distance, d metres, when travelling at a speed of v m/s is:

$$d = \frac{v^2}{2\mu g}$$

where μ is the coefficient of friction between the wheels and the road surface and g is gravitational force, where $g = 9.8 \text{ m/s}^2$.

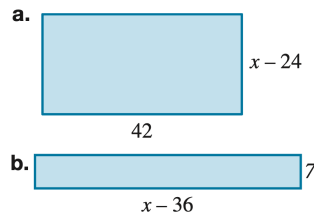
Calculate the stopping distances under the following conditions.

- (a) A car is travelling at a speed of 50 km/hr on a dry road with $\mu = 0.8$.
 - (b) Road and weather conditions, such as rain, change the value of μ .
 - i. If $\mu = 0.45$, find the stopping distance when travelling at a speed of 100 km/hr.
 - ii. If $\mu = 0.15$, find the stopping distance when travelling at a speed of 100 km/hr.
 - (c) Comment on your findings in part (b).
17. Two friends, Andrew and Anita, are enjoying an evening relaxing. They both have had 3 standard drinks in 2 hours and each weighs 70 kg. Discuss the difference in their blood alcohol content.
18. Andrew and Anita need to decide who can drive home. Both have P-plates. How long would they need to wait before driving, and who can legally drive first?

1.2 Equations

- 1. Three less than twice a number is 11. Form an equation and solve for the number.
- 2. A rectangle has a length 5 cm more than its width. If the perimeter is 38 cm, find the dimensions.
- 3. Your aunt from overseas sent an enormous box of chocolates for your mother's birthday. You were told not to eat chocolates. Nevertheless, sneaking into the pantry when your mother was out, you ate chocolates for five days, each day eating six more than the day before. After five days, you were caught out; it was discovered that you had eaten 100 chocolates, which made your mother extremely cross. How many chocolates had you eaten on each of the five days?
- 4. On her birthday today, a mother is three times as old as her daughter will be in eight years time. It just so happens that the mother's age today is the same as their house number, which is four times the value of thirteen minus the daughter's age.

- (a) Write each sentence above as an expression. (Hint: Make the daughter's age the unknown variable.)
 - (b) Use these two expressions to make an equation and solve it.
 - (c) How old are the mother and the daughter?
 - (d) What is the number of their house?
5. If the rectangles pictured at right have the same numerical value for their perimeter and area, calculate their dimensions.



1.3 Complex Calculations

1. Write an Excel formula to calculate the total cost if cell A1 contains quantity and cell B1 contains unit price.
2. In a spreadsheet, cell C3 contains the formula $=A3*B3$. If $A3 = 7$ and $B3 = 8$, what is displayed in C3?

2 Answers

2.1 Formulas

1. $A = 68$
2. $E = 225 \text{ J}$
3. $u = 10$
4. $h \approx 5.73$
5. Speed = 80 km/h
6. Distance = 30 km
7. Stopping distance = 58.8 m
8. (a) $BAC \approx 0.052$
(b) $H \approx 10.67$ hours
(c) Limitations: Does not account for individual metabolism, food intake, or type of alcohol.
9. Child dose $\approx 178.57 \text{ mg}$
10. Infant dose = 36 mg
11. $r = \sqrt{\frac{A}{\pi}}$
12. $u = \sqrt{v^2 - 2as}$
13. $x = \pm 3\sqrt{5}$
14. The student is correct. The company costs \$0.95 a kilometre.
15. (a) 12.30 metres (2 decimal places)
(b) i. 87.48 metres
ii. $d = \frac{(27.78)^2}{2 \times 0.15 \times 9.8} \approx 262.45 \text{ m.}$
(c) When μ (the coefficient of friction between the wheels and the road) is smaller, the stopping distance at the same speed is a lot longer.
16. Andrew's BAC is approximately 0.03 while Anita's BAC is approximately 0.04. While both had the same number of standard drinks in the same time and weigh the same, the difference comes from the denominators of their respective BAC formulae. The BAC formula for females will be dividing by a smaller number in the same weight, so it will give a higher BAC.
17. They would need to wait at least 2 hours and 8 minutes before Andrew drives home.

2.2 Equations

1. $2x - 3 = 11 \Rightarrow x = 7$
2. Width = 7 cm, Length = 12 cm
3. 8, 14, 20, 26, 32
4. (a)

$$\begin{aligned}\text{Mother's age} &= 3(d + 8) \\ \text{House number} &= 4(13 - d)\end{aligned}$$

(b)

$$\begin{aligned}3(d + 8) &= 4(13 - d) \\ 3d + 24 &= 52 - 4d \\ 3d + 4d &= 52 - 24 \\ 7d &= 28 \\ d &= 4\end{aligned}$$

(c)

$$\begin{aligned}\text{Mother:} & \quad 36 \text{ years old} \\ \text{Daughter:} & \quad 4 \text{ years old}\end{aligned}$$

(d) House number: 36

5. (a) $42 \text{ cm} \times 2.1 \text{ cm}$
(b) $7 \text{ cm} \times 2.8 \text{ cm}$

2.3 Complex Calculations

1. $=A1*B1$
2. 56